

An Abelian Tensor Stability Subcase for Seemingly Injective von Neumann Algebras

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Source and Question

The source is Gilles Pisier, *Seemingly injective von Neumann algebras*, arXiv:2010.13743. In Remark 1.7, Pisier asks whether tensor products preserve seeming injectivity:

particular p.p.p. is seemingly injective.

Remark 1.7. One major dissemblance with injectivity is that it is unclear (and a priori unlikely to be true) whether the tensor product of two seemingly injective algebras N, M is seemingly injective, even if $N = B(K)$ with K Hilbert. This is of course due to the fact that, unlike for c.p. maps, we cannot tensorize unital positive maps. Analogously, we do not know whether the commutant M' is seemingly injective if M is so.

In text, the question is whether the tensor product of two seemingly injective von Neumann algebras N, M is seemingly injective, even when $N = B(K)$. Pisier points out the obstruction: unlike completely positive maps, merely unital positive maps do not tensorize in general.

Definitions Used

Pisier calls a von Neumann algebra M *seemingly injective* if Id_M factors through some $B(H)$,

$$M \xrightarrow{u} B(H) \xrightarrow{v} M,$$

where u is normal, unital and positive, $\|v\|_{cb} = 1$, and $vu = \text{Id}_M$. Since v is unital and completely contractive, it is completely positive.

Pisier's Theorem 1.11 gives the equivalent matricial weak* positive approximation property: M is seemingly injective if and only if there is a net of normal finite-rank maps $T_\alpha = v_\alpha u_\alpha$ converging pointwise weak* to Id_M , where

$$M \xrightarrow{u_\alpha} M_{n(\alpha)} \xrightarrow{v_\alpha} M$$

has u_α, v_α unital positive and u_α normal.

Proof Intuition

The obstruction in Pisier's remark is real for noncommutative tensor factors: $\text{id}_{B(K)} \otimes u$ is positive only when u is completely positive. But if the other tensor leg is abelian, positivity can be checked coordinatewise. More precisely, approximate the abelian algebra by finite dimensional abelian algebras ℓ_∞^r . After tensoring with the matricial weak* PAP factorization for M , the intermediate

algebra is $\ell_\infty^r \otimes M_n$, a finite direct sum of matrix algebras. A block diagonal inclusion and a positive conditional expectation turn this into a legitimate matricial factorization. Thus the tensor product has Pisier's matricial weak* PAP, hence is seemingly injective.

Main Partial Result

Theorem 1. *Let A be an abelian von Neumann algebra and let M be seemingly injective. Then $A \overline{\otimes} M$ is seemingly injective.*

Proof. Since M is seemingly injective, Pisier's Theorem 1.11 provides a net

$$M \xrightarrow{u_\alpha} M_{n(\alpha)} \xrightarrow{v_\alpha} M$$

with u_α, v_α unital positive, u_α normal, and $T_\alpha = v_\alpha u_\alpha \rightarrow \text{Id}_M$ pointwise weak*.

Abelian von Neumann algebras are injective, hence semidiscrete. In the commutative case this gives a net of finite-rank normal unital positive maps

$$S_\beta = q_\beta p_\beta : A \rightarrow A$$

converging pointwise weak* to Id_A , with

$$A \xrightarrow{p_\beta} \ell_\infty^{r(\beta)} \xrightarrow{q_\beta} A$$

normal unital positive. Concretely, for $A = L^\infty(\Omega)$ these are the usual finite-partition averaging/simple-function maps.

For each pair (β, α) , define on $A \overline{\otimes} M$

$$R_{\beta, \alpha} := S_\beta \otimes T_\alpha.$$

We claim that $R_{\beta, \alpha}$ admits a matricial positive factorization. Indeed, $p_\beta \otimes u_\alpha$ maps $A \overline{\otimes} M$ into $\ell_\infty^{r(\beta)} \otimes M_{n(\alpha)}$. This tensor map is positive: if $p_\beta = (\varphi_1, \dots, \varphi_r)$ with positive normal coordinate functionals, then each coordinate is

$$(\varphi_i \otimes u_\alpha) = u_\alpha \circ (\varphi_i \otimes \text{id}_M),$$

a composition of positive maps. It is also normal and unital.

Likewise $q_\beta \otimes v_\alpha$ is positive on $\ell_\infty^r \otimes M_n$, because a positive element there is a tuple $(b_1, \dots, b_r)$ with $b_i \geq 0$, and it is sent to $\sum_i q_\beta(e_i) \otimes v_\alpha(b_i)$, a sum of positive tensors.

Now identify $\ell_\infty^r \otimes M_n$ with the block diagonal subalgebra of M_{rn} . Let $j : \ell_\infty^r \otimes M_n \rightarrow M_{rn}$ be the unital positive block inclusion and let $E : M_{rn} \rightarrow \ell_\infty^r \otimes M_n$ be the unital positive block-diagonal conditional expectation. Then

$$A \overline{\otimes} M \xrightarrow{j(p_\beta \otimes u_\alpha)} M_{rn} \xrightarrow{(q_\beta \otimes v_\alpha)E} A \overline{\otimes} M$$

is a matricial factorization through a matrix algebra by unital positive maps, with normal first leg, and its product is $S_\beta \otimes T_\alpha$.

The maps $S_\beta \otimes T_\alpha$ converge pointwise weak* to $\text{Id}_{A \overline{\otimes} M}$. This is immediate on algebraic simple tensors, and then follows for all of $A \overline{\otimes} M$ by uniform boundedness and weak* density of the algebraic tensor product. Hence $A \overline{\otimes} M$ has the matricial weak* PAP. Pisier's Theorem 1.11 gives that $A \overline{\otimes} M$ is seemingly injective. $\square$

A $B(K)$ -Tensor Subcase Under Finite Positivity

The abelian theorem above does not reach Pisier's explicit $B(K)$ frontier. There is, however, a clean noncommutative extension once the positive approximants on M have arbitrarily high finite positivity.

Theorem 2. *Let M be a QWEP von Neumann algebra. Suppose that for every integer $r \geq 1$ there is a net of normal finite-rank unital r -positive maps*

$$T_\alpha^{(r)} : M \rightarrow M$$

converging pointwise weak to Id_M . Then, for every Hilbert space K , the von Neumann algebra $B(K) \overline{\otimes} M$ is seemingly injective.*

Proof. By semidiscreteness of $B(K)$, or directly by compressing to finite dimensional subspaces of K , there is a net of normal finite-rank unital completely positive maps

$$S_\beta : B(K) \rightarrow B(K)$$

converging pointwise weak* to $\text{Id}_{B(K)}$, where each S_β factors through a finite dimensional C^* -algebra

$$F_\beta = \bigoplus_{j=1}^{m(\beta)} M_{r_j(\beta)}.$$

Write $R_\beta = \max_j r_j(\beta)$.

Choose, for each β , an approximating map $T_\alpha^{(R_\beta)}$ on M . We claim that

$$S_\beta \otimes T_\alpha^{(R_\beta)} : B(K) \overline{\otimes} M \rightarrow B(K) \overline{\otimes} M$$

is positive. Factor $S_\beta = q_\beta p_\beta$, with p_β and q_β unital completely positive. The map

$$p_\beta \otimes \text{id}_M : B(K) \overline{\otimes} M \rightarrow F_\beta \overline{\otimes} M$$

is completely positive, hence positive. On each summand $M_{r_j(\beta)} \overline{\otimes} M$, the map $\text{id}_{M_{r_j(\beta)}} \otimes T_\alpha^{(R_\beta)}$ is positive because $T_\alpha^{(R_\beta)}$ is R_β -positive and $r_j(\beta) \leq R_\beta$. Finally $q_\beta \otimes \text{id}_M$ is completely positive. The composition is exactly $S_\beta \otimes T_\alpha^{(R_\beta)}$, so it is a normal finite-rank unital positive map.

As (β, α) varies in the product directed set, these maps converge pointwise weak* to $\text{Id}_{B(K) \overline{\otimes} M}$. This is clear on algebraic simple tensors and then follows by uniform boundedness and weak* density.

It remains only to invoke Pisier's Theorem 1.11. The algebra $B(K) \overline{\otimes} M$ is QWEP: by Kirchberg's metric-surjection characterization of QWEP, tensoring a u.c.p. metric surjection onto M with the injective operator space $B(K)$ again gives a u.c.p. metric surjection onto the amplification $B(K) \overline{\otimes} M$. Thus $B(K) \overline{\otimes} M$ has the weak* PAP and is QWEP, hence is seemingly injective. $\square$

Corollary 1. *Let G be one of the groups for which the Haagerup-de Canniere multiplier construction cited in Pisier's Remark 1.12 gives, for every fixed $r \geq 1$, normal finite-rank unital r -positive multipliers converging pointwise weak* to the identity on $L(G)$. If $L(G)$ is QWEP, then*

$$B(K) \overline{\otimes} L(G)$$

is seemingly injective for every Hilbert space K . In particular this applies to the free-group factors covered by Pisier's discussion.

Proof. Apply Theorem 2 to the r -positive multiplier approximants. $\square$

Finite Matrix Companion

Corollary 2. *Let $n \geq 1$. Suppose M admits a seeming-injectivity factorization*

$$M \xrightarrow{u} B(H) \xrightarrow{v} M$$

with u normal, unital, positive and n -positive, and with v unital completely positive. Then $M_n \overline{\otimes} M$ is seemingly injective.

Proof. The map $\text{id}_{M_n} \otimes u : M_n \overline{\otimes} M \rightarrow M_n \overline{\otimes} B(H) \simeq B(\mathbb{C}^n \otimes H)$ is normal, unital and positive exactly because u is n -positive. The map $\text{id}_{M_n} \otimes v$ is unital completely positive, and

$$(\text{id}_{M_n} \otimes v)(\text{id}_{M_n} \otimes u) = \text{id}_{M_n \overline{\otimes} M}.$$

This is the defining factorization for seeming injectivity. □

Limitations

This does not solve the $B(K)$ case in Pisier’s remark for an arbitrary seemingly injective M . The proofs isolate the missing point. Either the finite-dimensional approximants on the other tensor leg must be abelian, or the approximants on M must be r -positive up to the needed matrix size. For a general seemingly injective algebra, Pisier’s theorem gives positive matricial weak* approximants, but not arbitrarily r -positive ones.

Novelty and Verification Notes

The local run indexes were searched for arXiv:2010.13743, “seemingly injective”, “remotely injective”, “tensor product”, “abelian”, “ $B(K)$ ”, “matricial weak* PAP”, and finite positivity variants. No duplicate packet was found. Web searches for exact phrases around “seemingly injective tensor product”, “seemingly injective abelian”, “seemingly injective $B(K)$ tensor”, and “seemingly injective remotely injective” did not locate an obvious later statement of these subcases. This is therefore marked as a likely-valid partial result pending human operator-algebra review, not as a claimed full solution of Pisier’s tensor-product question.

References

1. G. Pisier, *Seemingly injective von Neumann algebras*, arXiv:2010.13743.
2. A. Connes, *Classification of injective factors*, Ann. of Math. 104 (1976), 73–115.
3. M.-D. Choi and E. Effros, *Injectivity and operator spaces*, J. Funct. Anal. 24 (1977), 156–209.